

PRACTICAL - 10

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Section: 8

PART 1: INODE STRUCTURES, HARD LINKS AND SYMBOLIC LINKS

Aim

To investigate inode structures using commands such as `ls -li`, `stat`, and `find`, create hard links and symbolic links, and explain their impact on inode allocation.

Theory

An inode is a data structure in Linux that stores information about a file, such as file size, permissions, owner, timestamps, and location of data blocks.

Each file has an inode number.

- A hard link points to the same inode as the original file.
- A symbolic link has its own inode and stores the path of the original file.
- Creating a hard link does not create a new inode.
- Creating a symbolic link creates a new inode for the symbolic link.

Commands Used

Command	Purpose
<code>ls -li</code>	Displays inode number
<code>stat</code>	Displays detailed file information
<code>find</code>	Searches for files
<code>ln</code>	Creates a hard link
<code>ln -s</code>	Creates a symbolic link

Algorithm

1. Start.
2. Create a sample file.
3. Display its inode using `ls -li`.

4. Display file details using stat.
5. Create a hard link using ln.
4. Compare the inode numbers.
7. Create a symbolic link using ln -s.
8. Compare the inode numbers again.
9. Use find to locate the files.

10. Observe the inode allocation.

11. Stop.

Commands / Program

```
echo "Hello Linux" > original.txt
```

```
ls -li original.txt
```

```
stat original.txt
```

```
ln original.txt hardlink.txt
```

```
ls -li original.txt hardlink.txt
```

```
ln -s original.txt symlink.txt
```

```
ls -li original.txt hardlink.txt symlink.txt
```

```
stat original.txt
```

```
stat hardlink.txt
```

```
stat symlink.txt
```

```
find . -name "original.txt" -o -name "hardlink.txt" -o -name "symlink.txt"
```

Expected Observation

original.txt → Same inode

hardlink.txt → Same inode

symlink.txt → Different inode

The hard link and original file share the same inode, while the symbolic link has a separate inode.

Flow Chart

START



Create original.txt



Check inode using ls -li



Check details using stat



Create Hard Link



Compare inode numbers



Create Symbolic Link



Compare inode numbers



Use find command



Observe inode allocation



STOP

Output:

Result

The inode structure was successfully investigated using `ls -li`, `stat`, and `find`. It was observed that hard links share the same inode, while symbolic links have a separate inode.

PART 2: FILE READING AND WRITING USING `mmap()`

Aim

To design a program using `mmap()` to perform file reading and writing and compare memory-mapped I/O with traditional `read()/write()` operations in terms of efficiency and implementation complexity.

Theory

`mmap()` maps a file into the process's virtual memory. After mapping, the program can access the file directly through a memory pointer.

Traditional file I/O uses `read()` and `write()` system calls to transfer data between the file and a buffer.

Traditional I/O

`open()` → `read()` → modify → `write()` → `close()`

Memory-Mapped I/O

`open()` → `mmap()` → access memory → `msync()` → `munmap()` → `close()`

Memory-mapped I/O can be efficient for large files and repeated or random access because the program works directly with memory.

Algorithm

1. Start.
2. Open the file using `open()`.
3. Get the file size using `fstat()`.
4. Map the file into memory using `mmap()`.
5. Read the file contents through the mapped memory.

4. Modify the contents.
7. Synchronize changes using `msync()`.

8. Unmap the file using munmap().
- U. Close the file.
10. Stop.

C Program

- include <stdio.h>
- include <fcntl.h>
- include <unistd.h>
- include <sys/mman.h>
- include <sys/stat.h>
- include <string.h>

```
int main()
{
    int fd;
    struct stat st;
    char *data;
    fd = open("mmapfile.txt", O_RDWR); if
    (fd == -1)
    {
        perror("Error opening file");
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        perror("Error getting file size");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);

    if (data == MAP_FAILED)
    {
        perror("mmap failed");
        close(fd);
        return 1;
    }
}
```

```
printf("Original content:\n%s\n", data);
```

```
/* Write data using memory mapping */
```

```

    if (st.st_size >= 5)
    {
        memcpy(data, "HELLO", 5);
    }
    msync(data, st.st_size, MS_SYNC); printf("File
    modified successfully.\n"); munmap(data,
    st.st_size);
    close(fd);

    return 0;
}

```

Compilation

gcc mmap_program.c -o mmap_program

Execution

Create the input file:

echo "Hello Linux World" > mmapfile.txt Run:

./mmap_program

Check the file:

cat mmapfile.txt

Sample Output

Original content:

Hello Linux World

File modified successfully.

The file becomes:

HELLO Linux World

Comparison: mmap() vs read()/write()

Feature	read()/write()	mmap()
Method	Uses system calls	Maps file into memory
Efficiency	Good for sequential access	Efficient for repeated/random access
Implementation	Simple	More complex
Data access	Through buffers	Directly through memory
System calls	More read/write calls may be required	Fewer explicit I/O calls after mapping

Suitable for	Simple file operations	Large files and frequent access
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Flow Chart

START



Open File



Get File Size using fstat()



Map File using mmap()



Read File Through Memory



Modify File Data



msync()



munmap()



close()



STOP

Observation

The file was successfully read and modified using mmap(). Memory-mapped I/O provides direct memory-based access and can be efficient for large or repeatedly accessed files. However, it is slightly more complex than traditional read() and write() operations.

Output:

```

teja@Hasini:~$ nano mmapfile.txt
teja@Hasini:~$ ls
Experiment-2      parser.c
abc.c             path_resolver
destination.txt  path_resolver.c
dup2copy         pipeline
dup2copy.c       pipeline.c
filecopy         producer_consumer
filecopy.c       producer_consumer.c
filecopy.c.save  producer_consumer.c.save
filecopy.c.save.1 producer_consumer.c.save.1
fork             program.c
fork.c          question
hello           question.c
hello.c         question7
hello.txt       question7.c
input.txt       question9
linux          question9.c
merged.txt      quotes
mmapdemo.c      quotes.c
mmapfile.txt    readonly.txt
only_err.txt    redirection
only_out.txt    redirection.c
ordered.txt     redirection_pipeline
os_task         redirection_pipeline.c
os_task.c       result.txt
output.txt      sample.py
parser          source.txt
teja@Hasini:~$ gcc mmapdemo.c -o mmapdemo
teja@Hasini:~$ ./mmapdemo
Original content:
hello world

File modified successfully.
teja@Hasini:~$ cat mmapfile.txt
HELLO world

```

Result

The program successfully performed file reading and writing using mmap(). Memory-mapped I/O was compared with traditional read()/write() operations based on efficiency and implementation complexity.